

The Canonical Class of the MASA Context Stack and the Peres–Mermin Obstruction

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Abstract

Quantum measurements do not simply reveal pre-existing values: the outcome assigned to an observable can depend on which compatible observables are measured alongside it. This *contextuality* underlies the Peres–Mermin proof of the Kochen–Specker theorem and, in quantum computation with magic states, separates the classically simulable from the genuinely quantum — a separation known to turn on the parity of the local dimension. We give this even/odd divide a single geometric origin and a precise size. Organizing all measurement contexts of $M_n(\mathbb{C})$ into the stack $[X/\mathrm{PU}(n)]$, where X is the space of maximal abelian subalgebras, we show that the lifting class of $\mathrm{U}(1) \rightarrow \mathrm{U}(n) \rightarrow \mathrm{PU}(n)$ is vacuous on $\mathrm{U}(n) \ltimes X$ (Theorem 1) but, on $\mathrm{PU}(n) \ltimes X$, is a canonical class of *order exactly* n (Proposition 2, Theorem 8). Its real shadow, obtained from the signed-permutation subgroup $B_n = N(T) \cap O(n)$, is a $\mathbb{Z}_2$ -extension that splits if and only if n is odd (Theorem 3), with “ $-I$ is a rotation square iff n is even” as the mechanism — reproducing the qubit-contextual versus odd-qudit-noncontextual divide. At $n = 4$ it restricts to the real two-qubit Pauli class, whose pairing with the Peres–Mermin word is -1 (Theorem 6), and this pairing is exactly the obstruction to a noncontextual value assignment (Theorem 7). Thus the Peres–Mermin sign is not a combinatorial accident but the finite shadow of an identified, order- n cohomology class of the context stack.

1 Introduction and the vacuity theorem

Contextuality is the obstruction to assigning context-independent classical values to quantum observables. We organize *all* contexts of $M_n(\mathbb{C})$ into a single geometric object — the transformation groupoid of the conjugation action on the space of MASAs — and ask for the canonical cohomology class

that obstructs a global valuation. The naive guess pulls back the lifting class of the central extension $U(1) \rightarrow U(n) \xrightarrow{\pi} PU(n)$ along the arrow labels of $G = U(n) \ltimes X$. It is empty:

Theorem 1 (Vacuity). *Let $G = U(n) \ltimes X$ with arrows $(U, C): C \rightarrow UCU^\dagger$, and let ω be the groupoid 2-cocycle $\omega((V, \cdot), (U, \cdot)) = s(\bar{V})s(\bar{U})s(\overline{VU})^{-1}$ for a set-theoretic section s of π . Then $[\omega] = 0$ in $H^2(G; U(1))$.*

Proof. Each arrow carries an honest unitary. Set $\lambda(U, C) := U s(\bar{U})^{-1} \in U(1)$, well defined since U and $s(\bar{U})$ share an image in $PU(n)$. A direct computation gives $\omega = \delta\lambda^{-1}$; equivalently $(U, C) \mapsto (\lambda(U, C), (U, C))$ is a strict section of the extension groupoid. $\square$

The cocycle ω is genuinely asymmetric as a function on the group (a direct check finds it disagrees with its transpose on 120 of 256 pairs, so it is not a coboundary *there*); the point of Theorem 1 is that the extra arrow data of the *groupoid* trivializes it. The construction must therefore live on $\bar{G} := PU(n) \ltimes X$, over which the center acts trivially and the action genuinely factors.

2 The corrected object and Morita reduction

Fix the diagonal MASA $D \subset M_n(\mathbb{C})$. Its stabilizer in $U(n)$ is the normalizer of the maximal torus $N(T) = \{\text{monomial unitaries}\} \cong T \rtimes S_n$, and the isotropy of D in $\bar{G}$ is $PN(T) := N(T)/U(1)$.

Proposition 2 (Morita reduction). *The $PU(n)$ -action on $X \cong PU(n)/PN(T)$ is transitive, so $\bar{G}$ is Morita equivalent to the group $PN(T)$. Under this equivalence the canonical class $[\omega] \in H^2(\bar{G}; U(1))$ corresponds to the class of the central extension*

$$1 \longrightarrow U(1) \longrightarrow N(T) \longrightarrow PN(T) \longrightarrow 1. \quad (1)$$

Proof. The conjugation action of $PU(n)$ on the space X of MASAs is transitive (any MASA is $U(n)$ -conjugate to D), with stabilizer of D equal to $PN(T)$; hence $X \cong PU(n)/PN(T)$ as $PU(n)$ -spaces. For a transitive smooth action the space $PU(n)$ is a principal bibundle realizing a Morita equivalence $\bar{G} = PU(n) \ltimes X \simeq_{\text{Mor}} PN(T)$: $PU(n)$ carries a left $\bar{G}$ -action $(g, xPN(T)) \cdot p = gp$ over the moment map $p \mapsto pPN(T)$ and a commuting free, transitive right $PN(T)$ -action, and the two actions exhibit $PU(n)$

as a $\bar{G}$ - $PN(T)$ principal bibundle. Morita-equivalent groupoids have isomorphic stack cohomology with any coefficients [5, 6], so $H^\bullet(\bar{G}; \mathbb{U}(1)) \cong H^\bullet(PN(T); \mathbb{U}(1))$.

Under this isomorphism the canonical class $[\omega]$ maps to the restriction of the lifting datum to the isotropy. Indeed, ω is by construction the image of the lifting class of $\mathbb{U}(1) \rightarrow \mathbb{U}(n) \rightarrow \mathrm{PU}(n)$ under the classifying map $\bar{G} \rightarrow \mathrm{PU}(n)$ (arrow $\mapsto$ its $\mathrm{PU}(n)$ -label); pulling back along the bibundle restricts this classifying map to the inclusion $PN(T) \hookrightarrow \mathrm{PU}(n)$ on isotropy, and the pullback of the $\mathbb{U}(1)$ -extension $\mathbb{U}(n) \rightarrow \mathrm{PU}(n)$ along $PN(T) \hookrightarrow \mathrm{PU}(n)$ is exactly the extension $N(T) = \pi^{-1}(PN(T)) \rightarrow PN(T)$ of (1). Hence $[\omega]$ corresponds to the class of (1). $\square$

All subsequent content is a computation with the class of (1).

3 The real shadow: the even/odd dichotomy

Inside $N(T)$ sits the hyperoctahedral group of signed permutation matrices $B_n = \{\pm 1\}^n \rtimes S_n = N(T) \cap O(n)$, with $B_n \cap \mathbb{U}(1) \cdot I = \{\pm I\}$. Restricting (1) to $Q_n := B_n / \{\pm I\}$ yields the finite *real shadow*

$$1 \longrightarrow \mathbb{Z}_2 \longrightarrow B_n \longrightarrow Q_n \longrightarrow 1, \quad (2)$$

a $\mathbb{Z}_2$ -extension — the coefficient reduction $\mathbb{U}(1) \rightsquigarrow \mathbb{Z}_2$ that makes Section 5 possible.

Theorem 3 (Even/odd dichotomy). *The extension (2) splits if and only if n is odd.*

Proof. (n odd: splits.) The product-of-nonzero-entries map $\chi_s: B_n \rightarrow \{\pm 1\}$ is a homomorphism (nonzero entries of a product of monomial matrices multiply bijectively), and $\chi_s(-I) = (-1)^n = -1$, so $\ker \chi_s$ is a complement to $\{\pm I\}$. (n even: nonsplit.) With $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $g = J^{\oplus n/2} \in B_n$ one has $g^2 = -I$, so $-I$ is a *square* in B_n . Any complement has index 2 and is the kernel of a homomorphism $\chi: B_n \rightarrow \mathbb{Z}_2$ with $\chi(-I) = -1$; but homomorphisms to $\mathbb{Z}_2$ kill squares. Contradiction. $\square$

(Verified computationally for $n = 2, \dots, 12$.) The dichotomy is geometric: $-I$ is a rotation square exactly in even dimensions.

4 The real Pauli slice and gauge invariance

Take $n = 4$, $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$. Since $Y = iXZ$, every two-qubit Pauli operator is, up to phase, a real signed permutation matrix. Let $R = \{\pm A \otimes B : A, B \in \{I, X, Z, XZ\}\} \subset B_4$, $|R| = 32$, with quotient $\bar{P} := R/\{\pm I\} \cong \mathbb{Z}_2^4$ the projective two-qubit Pauli group.

Theorem 4 (Real Pauli slice). *The preimage of $\bar{P} \subset Q_4$ under $B_4 \rightarrow Q_4$ is exactly R . Hence the shadow class restricts to the class $[\beta_R]$ of $1 \rightarrow \mathbb{Z}_2 \rightarrow R \rightarrow \mathbb{Z}_2^4 \rightarrow 1$, whose commutator pairing is the standard symplectic form: $[\tilde{u}, \tilde{v}] = (-1)^{\langle u, v \rangle} I$, $\langle u, v \rangle = \sum_{\text{qubits}} (a_i d'_i + b_i c'_i)$.*

(Verified exhaustively: preimage = R over all 384 elements of B_4 ; commutator = $(-1)^{\langle u, v \rangle}$ over all 256 pairs.)

Lemma 5 (Gauge-invariance criterion). *Let $1 \rightarrow A \rightarrow E \rightarrow Q \rightarrow 1$ be central and $w = g_k \cdots g_1$ a word with $w = 1$ in Q . The element of A obtained by lifting each letter arbitrarily and multiplying in E is independent of the lifts iff, for every letter g , its total exponent in w annihilates A ; for $A = \mathbb{Z}_2$, iff every letter occurs an even number of times.*

Proof. Relifting g by $a \in A$ multiplies the evaluation by $a^{e(g)}$; centrality makes the contributions independent. $\square$

This is the mod-2 Hopf pairing in disguise: even-multiplicity words represent $\mathbb{Z}_2$ -2-cycles. Consequently the $U(1)$ -level extension (1) attaches an invariant only to exponent-sum-zero words (commutators), which already prove $[\beta_R] \neq 0$ via the symplectic form; the Mermin word is not exponent-zero, so its sign is *not* a $U(1)$ -level invariant. In the $\mathbb{Z}_2$ shadow — available precisely because of the real structure — even multiplicity suffices, and the Mermin word qualifies.

5 Detection and the bridge to valuations

Arrange the nine Peres–Mermin observables

$$\begin{array}{ccc} X \otimes I & I \otimes X & X \otimes X \\ I \otimes Y & Y \otimes I & Y \otimes Y \\ X \otimes Y & Y \otimes X & Z \otimes Z \end{array}$$

Hermitian constraint signs: all three rows and the first two columns give $+I$; the third column $\{X \otimes X, Y \otimes Y, Z \otimes Z\}$ gives $-I$. Let μ_{PM} be the 18-letter

concatenation of the six context words; each letter occurs exactly twice, so by Lemma 5 its evaluation in R over arbitrary $\pm$ real lifts is a well-defined element of $\{\pm I\}$.

Theorem 6 (Detection). $\langle [\beta_R], \mu_{\text{PM}} \rangle = -1$. Hence $[\beta_R] \neq 0$, the shadow class (2) is nonzero at $n = 4$, the extension (1) is nonsplit, and the canonical class $[\omega] \in H^2(\text{PU}(4) \ltimes X; \text{U}(1))$ of the MASA context stack is nontrivial, witnessed by the Peres–Mermin configuration.

(Evaluated over all $2^9 = 512$ letter relifts: $-I$ in every case.)

Theorem 7 (Bridge). A noncontextual value assignment is a map v from the nine observables to $\{\pm 1\}$ with $\prod_{O \in c} v(O) = \epsilon(c)$ for each context c . Such v exists iff $\prod_c \epsilon(c) = +1$; since each observable lies in exactly two contexts, this product equals the pairing of Theorem 6, $= -1$, so no assignment exists, and nonexistence is equivalent to nonvanishing of that pairing.

Proof. Multiplying the six constraints leaves $\prod_c \epsilon(c)$ on the right and $\prod_O v(O)^2 = +1$ on the left. Exhaustive search finds 0 of the 2^9 assignments satisfy all six constraints, matching $\prod \epsilon = -1$. $\square$

The obstruction is thus not a “loop through six contexts”: every projective Pauli fixes all six PM MASAs, so the configuration sits inside the isotropy, and the correct cycle is a 2-cycle there (Lemma 5), with the six contexts entering as the cover over which trivialization is demanded.

6 Two-tier invariance and consistency with physics

Two coefficient levels carry different invariants: the $\text{U}(1)$ class (1) detects the symplectic form via commutators (nonsplitness) but not the Mermin sign; the $\mathbb{Z}_2$ shadow (2) carries the Mermin sign itself, via even-multiplicity words. The reduction is furnished geometrically by $N(T) \cap O(n) = B_n$. For even n the shadow is nonsplit and, at $n = 4$, is Peres–Mermin — matching the contextuality of the qubit stabilizer subtheory. For odd n the shadow splits (χ_s), consistent with Gross’s theorem and the nonnegative discrete Wigner function for odd-prime-dimensional stabilizer subtheories [3, 2].

7 The order of the canonical class

Theorem 8 (Order). The class of (1) — equivalently the canonical class $[\omega]$ of $[X/\text{PU}(n)]$ — has order exactly n in $H^2(\text{PN}(T); \text{U}(1))$.

Proof. Upper bound $n[\omega] = 0$. The n -fold Baer pushout is $E_n = (\mathrm{U}(1) \times N(T))/\{(z, g) \sim (\lambda^n z, \lambda^{-1} g)\}$, and $\sigma(\bar{g}) := [(\det g)^{-1}, g]$ is a well-defined homomorphic section (as $\det(\lambda g) = \lambda^n \det g$). This determinant twist splits the n -th power — the group-level shadow of the n -torsion of Dixmier–Douady classes of $\mathrm{PU}(n)$ -bundles.

Lower bound (no $m < n$ works). Restrict to the scalar μ_n -shadow $\overline{\mu_n \wr S_n}$. A splitting of the m -th pushout is an S_n -equivariant retraction of $1 \rightarrow \mathbb{Z}_n \rightarrow D \rightarrow D/Z \rightarrow 1$ ($D = \mathbb{Z}_n^n$, Z the scalar diagonal), i.e. an S_n -equivariant homomorphism $\varphi: \mathbb{Z}_n^n \rightarrow \mathbb{Z}_n$ with $\varphi(\mathrm{diag}) \equiv m$. By the Lemma below $\varphi(x) = k \sum_i x_i$, so on the diagonal generator $\varphi(1, \dots, 1) = kn \equiv 0 \pmod{n}$ for every k . Hence the m -pushout splits iff $\exists k: kn \equiv m \pmod{n}$, i.e. iff $n \mid m$. The smallest positive such m is n . $\square$

Lemma 9. *The only S_n -equivariant homomorphisms $\mathbb{Z}_n^n \rightarrow \mathbb{Z}_n$ are $x \mapsto k \sum_i x_i$, $k \in \mathbb{Z}_n$ (exactly n of them).*

Proof. An additive map is $x \mapsto \sum_i c_i x_i$ with $c_i = \varphi(e_i)$. Since S_n permutes the e_i transitively, equivariance forces all c_i equal to a single k . $\square$

The proof is dimension-independent; Lemma 9 is verified to give exactly n equivariant maps for $n = 2, \dots, 8$, and the order is confirmed $= n$ by direct equivariant-section search over the same range. In the LHS spectral sequence for $PN(T) = (T/\mathrm{U}(1)) \rtimes S_n$ the class saturates the equivariance-obstruction component $E_2^{1,1} = H^1(S_n; \hat{T}') \cong \mathbb{Z}/n$; the permutation-copy of S_n splits the extension, so the class has no $E_2^{2,0} = H^2(S_n; \mathrm{U}(1)) = \mathbb{Z}_2$ component.

Proposition 10 (The full second cohomology). *For $n \geq 4$, $H^2(PN(T); \mathrm{U}(1)) \cong \mathbb{Z}/n \oplus \mathbb{Z}_2$, and the canonical class generates the $\mathbb{Z}/n$ summand.*

Proof. In the LHS sequence for $PN(T) = (T/\mathrm{U}(1)) \rtimes S_n$ the total-degree-2 entries are $E_2^{0,2} = H^0(S_n; H_{\mathrm{sm}}^2(T/\mathrm{U}(1); \mathrm{U}(1))) = 0$ (smooth $\mathrm{U}(1)$ -central extensions of a torus split), $E_2^{1,1} = H^1(S_n; \hat{T}') = \mathbb{Z}/n$, and $E_2^{2,0} = H^2(S_n; \mathrm{U}(1)) = \mathbb{Z}_2$ (the Schur multiplier, $n \geq 4$). The only differential into $E_2^{2,0}$ is $d_2: E_2^{0,1} \rightarrow E_2^{2,0}$, and $E_2^{0,1} = (\hat{T}')^{S_n} = (A_{n-1})^{S_n} = 0$ (the S_n -invariants of the root lattice vanish: an invariant vector is a multiple of $(1, \dots, 1)$, which has nonzero coordinate sum), so $E_2^{2,0} = \mathbb{Z}_2$. Nothing maps into $E_2^{1,1}$ ($E_2^{1-r,r} = 0$ for $r \geq 2$); its only possible outgoing differential is $d_2: E_2^{1,1} \rightarrow E_2^{3,0}$. By Theorem 8 there is a class of order exactly n lying, by the splitting just noted, in filtration level 1, so $E_\infty^{1,1}$ contains an element of order n ; being a subquotient of $\mathbb{Z}/n$ it equals $\mathbb{Z}/n$, and $d_2|_{E^{1,1}} = 0$. The degree-2 filtration is

thus $0 \rightarrow \mathbb{Z}_2 \rightarrow H^2 \rightarrow \mathbb{Z}/n \rightarrow 0$. The canonical class c has order n and projects to a generator of the quotient, so $\langle c \rangle \cong \mathbb{Z}/n$ is a complement to $\mathbb{Z}_2$; the extension splits and $H^2(PN(T); \mathbb{U}(1)) = \mathbb{Z}/n \oplus \mathbb{Z}_2$, with c generating $\mathbb{Z}/n$. $\square$

8 The qudit shadow dichotomy and cycle arithmetic

Replacing B_n by the d -monomial shadow $\mu_d \wr S_n = \mathbb{Z}_d^n \rtimes S_n$ (kernel the scalar diagonal $\mathbb{Z}_d$) gives, by the same equivariant-retraction arithmetic:

Theorem 11 (Qudit dichotomy). *The shadow extension $1 \rightarrow \mathbb{Z}_d \rightarrow \mu_d \wr S_n \rightarrow Q_{n,d} \rightarrow 1$ splits iff $\gcd(n, d) = 1$; the m -th Baer multiple splits iff $\gcd(n, d) \mid m$, so the shadow class has order exactly $\gcd(n, d)$, and the same holds at the $\mathbb{U}(1)$ level.*

Theorem 3 is $d = 2$: $\gcd(n, 2) = 1$ iff n odd. A cycle-arithmetic companion controls which configurations convert the twist into a Kochen–Specker obstruction:

Theorem 12 (Cycle arithmetic). *In a central extension with kernel $\mathbb{Z}_d$, a word in which every letter occurs exactly r times has lift-independent evaluation iff $d \mid r$. Mermin-square configurations (each observable in exactly two contexts) therefore carry a well-defined obstruction iff $d \mid 2$.*

The qutrit Heisenberg group ($\gcd(3, 3) = 3$, nonsplit) has no admissible multiplicity-2 cycle ($3 \nmid 2$), recovering odd- d noncontextuality as cycle arithmetic. Peres–Mermin is the unique intersection $n = 4, d = 2, r = 2$: shadow nonzero ($\gcd(4, 2) = 2$), cycles admissible ($2 \mid 2$), pairing -1 . (All splitting predictions verified across (n, d, m) ; see appendix.)

9 Related work

The obstruction of the two-qubit Pauli extension over $\mathbb{Z}_2^4$ to PM value assignments, and the symplectic commutator form, are classical and appear in the cohomological contextuality program of Okay–Roberts–Bartlett–Raussendorf [1]; Theorem 7 is their statement in our normalization. The even/odd HW dichotomy and odd- d noncontextuality are due to Gross and Raussendorf et al. [3, 2]. A recent preprint [4] proposes a categorified spectral-stack view of operator systems and, by way of illustration, discusses

Mermin–Peres contextuality as a descent obstruction; its H^2 obstruction class is deferred to a stated companion. We note it as conceptually adjacent but do not rely on it, and we make no claim about its detailed content: our construction and all results here are independent. Our contribution is specific and self-contained: (i) the obstructing class is *identified* as the Morita reduction of the $U(1) \rightarrow U(n) \rightarrow PU(n)$ lifting class over $[X/PU(n)]$ (Prop. 2, Thm. 6); (ii) it has order exactly n (Thm. 8); (iii) its real-shadow arithmetic yields the qubit/odd-qudit dichotomy (Thm. 3, 11)

10 Open problems

(1) Does the canonical class see non-parity contextuality (Yu–Oh at $n = 3$, where the real shadow splits)? Either a non-scalar-monomial subgroup carries it, or a scope no-go holds. (2) A groupoid-cohomological Arason–Vinberg–Nagel-type bridge generalizing Thm 7; compare Carù before claiming completeness. (3) Realize the class as a Dixmier–Douady gerbe on $[X/PU(n)]$ and compute its H^3 image.

A Computational certificates

`verify_chain.py`: PM constraint signs (Hermitian and real); $-I$ square analysis in B_n , $n \leq 14$; $|R| = 32$, $R \subset B_4$, preimage of $\bar{P} = R$ over all 384 elements; commutator = symplectic form over 256 pairs; Mermin word = $-I$ over all 512 relifts; 0 noncontextual assignments among 2^9 , $\prod_c \epsilon(c) = -1$. `verify_part2.py`: Theorem 11 splitting predictions across (n, d, m) ; qutrit Heisenberg order 27; cycle arithmetic. `thmG_general.py`: Lemma 9 (n equivariant homs, $n=2..8$) and order = n ($n=2..8$). `convention_independence.py`: certificate value invariant across four Weyl normalizations.

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